

The exponential function

①

Let $z = x + iy$. We define $e^z = e^x e^{iy}$ & where the symbol e^{iy} stands for $\cos y + i \sin y$. (y in radians).

Properties of exponential in $\exp(z)$ or e^z

1) $e^{z_1} e^{z_2} = e^{z_1 + z_2} \Rightarrow \frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$

2) $e^{z + 2\pi i} = e^z$

3) $|e^z| = e^x$ & $\arg(e^z) = \{y + 2\pi n | n \in \mathbb{Z}\}$

4) $e^z \neq 0$ for any complex no z .

If $e^z = 0$ for some $z \in \mathbb{C}$, then, $e^z = 0 \rightarrow 1 = 0$

Properties not possessed by $f(x) = e^x$ $x \in \mathbb{R}$.

5) (Periodicity) $e^{2\pi i + z} = e^z$

e^z is periodic with a purely imaginary period of $2\pi i$

6) e^z can be negative even though e^x is always pos
(Eg $z = \pi i$)

7) $\mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$ is onto $\mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$
 $z \rightarrow e^z$

i.e. $\forall w_0 \neq 0$ in $\mathbb{C}$, $\exists z_0 \in \mathbb{C}$ st $e^{z_0} = w_0$.

Recall $\mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$ is not onto under e^x

but $\mathbb{R} \xrightarrow{\exp} \mathbb{R}_{>0}$ is onto

$x \rightarrow e^x$

$\forall y_0 > 0$, $\exists x_0 \in \mathbb{R}$ st $e^{x_0} = y_0$. x_0 is what we call as $\ln y_0$.

$\ln y_0$.

Pr 1) Find all $z = x+iy$ for which ~~$e^z = 1+i$~~ $e^z = 1+\sqrt{3}i$

Soln: $e^{x+iy} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2e^{i\frac{\pi}{3}}$

$\therefore$ ~~$e^x = 2$~~ $e^x = 2$ & $y \in \left\{\frac{\pi}{3} + 2n\pi \mid n \in \mathbb{Z}\right\}$

ie, $x = \ln 2$ & y can be any no in the set above

$\therefore z = \ln 2 + i\left(\frac{\pi}{3} + 2n\pi\right)$ for any $n \in \mathbb{Z}$

satisfies $e^z = 1+\sqrt{3}i$

Problems in exponential function

Churchill Pg 91 Example

Pg 92: 1, 3, 5, 7, 8a, 8c, 10

Logarithmic functions

(only applicable if $z > 0$)

Recall that $\ln x$ is that unique real no. y such that $e^y = x$

Similarly, here we seek for all $w \in \mathbb{C}$ such that $e^w = z$ — (1)

& would like to define these w as candidates for

$\log z$.

Let us write $z = re^{i\theta}$ where $-\pi < \theta \leq \pi, z \neq 0$. Note that if w is a soln for (1), then so is $w + 2\pi ni$ for any $n \in \mathbb{Z}$ as $e^{w+2\pi ni} = e^w = z$.

Hence, $\log z$ if exists is a multi-valued function.

Now, if $w = u + iv$, then (1) is eqvt to saying

$$e^{u+iv} = re^{i\theta}$$

That is, $e^u = r$ & v is of the form $\theta + 2\pi n$ for any $n \in \mathbb{Z}$.

ie, $u = \ln r$ & $v \in \{\theta + 2\pi n, n \in \mathbb{Z}\}$.

Hence, $e^w = z$ iff

$$w = \ln r + i(\theta + 2\pi n), n \in \mathbb{Z}.$$

$$w = \ln r + i\theta + 2\pi ni, n \in \mathbb{Z} \rightarrow (1)$$

But $\theta + 2\pi n, n \in \mathbb{Z}$ are the values of $\arg z$.

Definition:- Let $z \neq 0 \in \mathbb{C}$. Then,

$$\log z \stackrel{\text{defn}}{=} \ln r + i \arg z$$

$$\text{Obsn 1: } e^{\log z} = e^{\ln r} e^{i \arg z} = re^{i\theta} = z$$

$$\text{Obsn 2: } e^z = e^x \cdot e^{iy} =$$

Comparing the polar forms, we get $|e^z| = e^x$ & $\arg(e^z) = y + 2\pi n$

$n \in \mathbb{Z}$.

$$\Rightarrow \log(e^z) = \ln e^x + i \arg(e^z)$$

$$= x + i(y + 2\pi n)$$

$$= z + 2\pi ni, n \in \mathbb{Z} \rightarrow (2)$$

Comparison between real logarithm & complex logarithm

Real variable $\ln x$

$$\log(e^x) = x \quad \checkmark$$

$$e^{\log x} = x \quad \checkmark$$

Complex logarithm

$$\log(e^z) = z \text{ is false (see (2))}$$

$$e^{\log z} = z$$

$\exp(z)$ is single valued although $\log z$ isn't.

Pr 1) Find $\log(-1 - \sqrt{3}i)$

$$\text{Let } z = -1 - \sqrt{3}i = 2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 2e^{i(-2\pi/3)}$$

$$\therefore \log z = \ln 2 + i(-\frac{2\pi}{3} + 2\pi n), n \in \mathbb{Z}$$

Principal value of the logarithm

To avoid the ambiguities when multi-valued functions arise, we need to define a single valued logarithm function (like in case of argument function).

Define $\rightarrow$ Capital L

$$\text{Log } z = \ln |z| + i \text{Arg } z$$

Further, if $z = re^{i\theta}$ where $-\pi < \theta \leq \pi$, then

$$\log z = \ln r + i\theta \quad (\text{ie, } n=0 \text{ in (1)})$$

$$\therefore \log z = \text{Log } z + 2\pi ni$$

Further if $z \in \mathbb{R}_{>0}$, then $z = re^{i\theta}$ (where $\theta = 0$ & hence $z = r$) $\Rightarrow \log z = \ln r$

Thus, $\text{Log } z$, the principal value of the logarithm is the extension of the usual real (natural) logarithm.

Continuity of the $\text{Arg } z$ (principal argument)

$\text{Arg } z$ is defined on $\mathbb{C} \setminus \{0\}$.

It is continuous at every point in $\mathbb{C}$ except at zero & on the negative real axis.

To see this, consider the polar form of z ie, $re^{i\theta}$ & write $\text{Arg}(z)$ as a function $f: \mathbb{R}^+ \times \mathbb{R} \rightarrow (-\pi, \pi]$

$$f(r, \theta) = \text{Arg}(re^{i\theta})$$

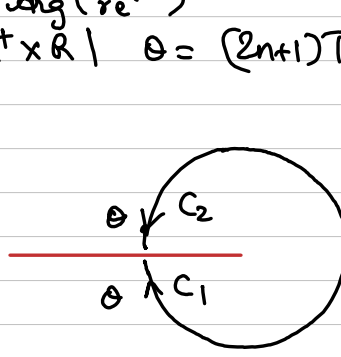
$$\text{Let } S = \{(r, \theta) \in \mathbb{R}^+ \times \mathbb{R} \mid \theta = (2n+1)\pi, n \in \mathbb{Z}\}$$

for simplicity, consider

$$(r, \pi) \in S.$$

This can be generalised

easily to any $(r, (2n+1)\pi) \in S$.



To show discontinuity at a point, it is enough to find two curves that approach (r, π) but give different limits.

C_1 & C_2

$$\lim_{\theta \rightarrow \pi^+} f(r, \theta) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \text{Arg}(re^{i(\pi+h)}) = \lim_{h \rightarrow 0^+} \pi + h - 2\pi = -\pi.$$

$$(r, \pi) \nwarrow \begin{matrix} C_2 \\ C_1 \end{matrix} (r, \theta = \pi + h)$$

$$\lim_{\theta \rightarrow \pi^-} f(r, \theta) = \lim_{h \rightarrow 0^+} \text{Arg}(re^{i(\pi-h)}) = \lim_{h \rightarrow 0^+} \pi - h = \pi$$

$$C_2 \nearrow \begin{matrix} (r, \pi-h) \\ (r, \pi) \end{matrix}$$

At other points, ie, $\mathbb{C} \setminus S$, it is easily seen that $f(r, \theta)$ is continuous.

Conclusion:- $\log z$ is cont. at $z \Leftrightarrow \text{Arg } z$ is cont. at z .

Theorem (Sufficient condition for differentiability in terms of polar form)

Let the function $f(z) = u(r, \theta) + iv(r, \theta)$ be defined throughout some ε -nbhd of a non-zero point $z_0 = r_0 e^{i\theta_0}$ & suppose that

a) the first order partial derivatives of the functions u & v with respect to r & θ exist everywhere in the nbhd

b) those partial derivatives are continuous at (r_0, θ_0) & satisfy the polar form of the CE eqns

$$ru_r = v_\theta$$

$$u_\theta = -rv_r \quad \text{at } (r_0, \theta_0).$$

Then,

$\left(\frac{df}{dz}\right)$ exists at $z = z_0$ & its value is

$$f'(z_0) = e^{-i\theta_0} (u_r + iv_r) \Big|_{(r_0, \theta_0)}.$$

Derivative of the logarithm

Consider $r \in \mathbb{R}^+$ & $\theta \in \mathbb{R}$. Then, $\log(re^{i\theta})$ is defined as

$$\log(re^{i\theta}) = \ln r + i \text{Arg}(re^{i\theta})$$

$$u(r, \theta) = \ln r \quad \& \quad v(r, \theta) = \text{Arg}(re^{i\theta}).$$

Remark:- Note that if $\theta \in (-\pi, \pi]$, then

$$\log(re^{i\theta}) = \ln r + i\theta$$

$$\text{with } u(r, \theta) = \ln r \quad \& \quad v(r, \theta) = \theta.$$

Going back to eqn (3), for any $(r_0, \theta_0) \in \mathbb{R}^+ \times \mathbb{R}$,

$$u_r(r_0, \theta_0) = \frac{1}{r_0}, u_\theta(r_0, \theta_0) = 0$$

$$v_r(r_0, \theta_0) = 0$$

However, $\frac{\partial v}{\partial \theta}(r_0, \theta_0)$ doesn't exist if $(r_0, \theta_0) \in S$.

$$\lim_{\theta \rightarrow \pi^+} \frac{v(r_0, \theta) - v(r_0, \pi)}{\theta - \pi} \quad \text{But } v(r_0, \theta) = \pi + h - 2\pi = -\pi + h$$

$$= \lim_{h \rightarrow 0^+} \frac{-\pi + h - \pi}{h} = \text{doesn't exist.}$$

On the other hand, the left hand derivative

$$\lim_{\theta \rightarrow \pi^-} \frac{v(r_0, \theta) - v(r_0, \pi)}{\theta - \pi} = \lim_{h \rightarrow 0^+} \frac{(\pi - h) - \pi}{-h} = 1$$

Hence, $\frac{\partial v}{\partial \theta}(r_0, \theta_0)$ doesn't exist if $\theta_0 \in \{(2n+1)\pi \mid n \in \mathbb{Z}\}$.

Check: $\frac{\partial v}{\partial \theta}(r_0, \theta_0) = 1$ if $(r_0, \theta_0) \in \mathbb{C} \setminus S$.

(Note that $\text{Arg}(re^{i\theta})$ equals θ upto a difference factor which is a multiple of 2π)

$$\therefore \frac{\partial v}{\partial \theta} = \frac{\partial(\theta - 2\pi n)}{\partial \theta} = 1. \text{ if } \frac{\partial v}{\partial \theta} \text{ exists.}$$

One can now easily verify that all the conditions of Theorem above are met if we define $\log z$ for $z \in \mathbb{C} \setminus S$. Thus, its derivative exists for every $z \in \mathbb{C} \setminus S$ & given by

$$f'(z_0) = \left[e^{-i\theta} (u_r + iv_r) \right] \Big|_{(r_0, \theta_0)} \rightarrow z_0 = r_0 e^{i\theta_0}$$

$$u_r(r_0, \theta_0) = \frac{1}{r_0}$$

$$v_r(r_0, \theta_0) = 0$$

$$\Rightarrow f'(z_0) = e^{-i\theta_0} \times \frac{1}{r_0} = \frac{1}{r_0 e^{i\theta_0}} = \frac{1}{z_0}$$

$\therefore$ If $z_0 \in \mathbb{C} \setminus S$, then

$$\left[\frac{d(\log z)}{dz} \right](z_0) = \frac{1}{z_0}.$$

Summary:-

1) $\log z$ is defined on $\mathbb{C} \setminus \{0\}$.

2) $\log z$ is not continuous on $\mathbb{R}_{\leq 0}$.

($\text{Arg } z$)

3) $\log z$ is analytic on $\mathbb{C} \setminus S$ &

its derivative is $\frac{1}{z}$.

Corollary:- $\nexists$ no open ball $B(0, \rho)$ centered around 0 with radius ρ where $\log z$ is analytic

Hence, $\nexists$ no simply connected domain containing origin where $\log z$ is analytic.

Hence, $\frac{1}{z}$ will never have an antiderivative on any simply connected domain containing origin.

Recall: If C = unit circle with CCW orientation, then

$$\oint_C \left(\frac{1}{z}\right) dz = 2\pi i$$

This remark also tells you that $\frac{1}{z}$ doesn't have an antiderivative on any SCD that contains C inside it.

Identities involving $\log$

(Note that for real $x_1 > 0, x_2 > 0$,

$$\log(x_1 x_2) = \log x_1 + \log x_2$$

where $\log x$ is single valued.)

$$1. \log(z_1 z_2) = \log z_1 + \log z_2$$

However, $\log(z_1 z_2) \neq \log z_1 + \log z_2$.

Pr 1) ~~What is the value of $\log z$~~ Show that when
 $z_1 = i$ & $z_2 = \frac{-1-i}{\sqrt{2}}$,

$$\log(z_1 z_2) \neq \log z_1 + \log z_2$$

$$\log z_1 = \log(e^{i\pi/2}) = i\pi/2.$$

$$\log z_2 = \log(e^{i3\pi/4}) = i3\pi/4.$$

$$\begin{aligned} \log(z_1 z_2) &= \log\left(i \left(\frac{-1-i}{\sqrt{2}}\right)\right) = \log\left(\frac{1-i}{\sqrt{2}}\right) = \log(e^{-i\pi/4}) \\ &= -i\pi/4. \end{aligned}$$

hence, proved.

Ex Pr 2) Show that if $\operatorname{Re}(z_1) > 0$ & $\operatorname{Re}(z_2) > 0$, then

$$\log(z_1 z_2) = \log z_1 + \log z_2$$

$$\operatorname{Re}(z_j) > 0 \Rightarrow \log z_j = \ln|z_j| + i \operatorname{Arg} z_j \quad \text{where } \operatorname{Arg} z_j \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\text{Hence, } z_1 z_2 = r_1 r_2 e^{i(\operatorname{Arg} z_1 + \operatorname{Arg} z_2)}$$

$$\text{Since } \operatorname{Arg} z_1 + \operatorname{Arg} z_2 \in (-\pi, \pi], \operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2 \text{ \&}$$

$$\text{hence, } \cancel{z_1 z_2 = r_1 r_2 e^{i \operatorname{Arg} z_1 z_2}}$$

$$\log z_1 z_2 = \ln|z_1 z_2| + i(\operatorname{Arg} z_1 + \operatorname{Arg} z_2)$$

$$= \ln|z_1| + \ln|z_2| + "$$

$$= \log z_1 + \log z_2.$$